Kolanti Avinash

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.00
Narayana Junior College	Class XII (Board Of Intermediate Education, Andhra Pradesh)	2021	97.40%
Narayana EM High School	Class X (Board Of Secondary Education, Andhra Pradesh)	2019	10.00%

SKILLS

Simulation Tools: Cadence Virtuoso, LT Spice, MATLAB, Xilinx Vivado.

Programming Languages: Verilog, C

RELEVANT COURSES

Analog Domain: Network Theory, Analog Electronics, Analog IC Design, Analog IC design lab.

Digital Domain: Digital Electronics, Digital IC design lab, CMOS Digital Design.

Others: Semiconductor devices and modelling, Memory Devices for AI/ML Computing.

THESIS

Instructor : Dr. Abhishek Kumar

Design of transceiver for On board Wireless interface of distributed systems.

- Designing a Low Noise Amplifier(LNA) for the receiver side in cadence virtuoso (TSMC 65nm tech node).
- And inductor less CMOS CS Amplifier with resistive feedback for the wideband input matching.
- It forms self biased inverter based amplifier which offers high gain and low noise at low power consumption.
- With Electrical characteristics of (VDD =1.2v, 2.45GHz and 50 ohm system).
- And achieved small signal gain (S21) = 15.4 dB and Input return loss (S11)=-21 dB.
- And further will work on folded Cascode OTA.

PROJECTS

Two-Stage Operational Transconductance Amplifier (OTA) | *Course Project*

Course : Analog IC Design Lab | Instructor : Prof. Ashudeb Dutta

- Designed a Two stage single ended OTA using GPDK 45nm technology node in Cadence Virtuoso.
- Considering VDD = 1.8 V and performed DC, AC, Transient analysis and slew-rate to evaluate OTA performance.
- Using the given specifications we Achieved DC Gain = 65 dB, UGB = 87 MHz and Phase Margin of 76°.

Band Gap Reference Circuit | *Course Project*

Course : Analog IC Design Lab | Instructor : Prof. Ashudeb Dutta.

- Designed an OTA - based Based Band Gap Reference Circuit using GPDK 45nm tech node in cadence virtuoso.
- It will Generate a Constant reference Voltage independent of Temperature and Supply variations.
- Designed and optimized current mirror circuits and OTA.
- Generated a Reference voltage of VREF = 1.17 V using PTAT/CTAT compensation.

- Verified VREF across all the PVT corners and temperature range of 40°C to 125°C.
- Achieved the stable VREF of 1.17 V.

Low-Dropout Regulator

- Designed a LDO in cadence virtuoso using TSMC 28nm Technology node.
- Using PMOS two stage OTA coupled with a PMOS pass transistor including transistor sizing and feedback-network design.
- Performed loop stability and AC analysis, achieving 66 dB loop gain, 68.5° phase margin, and 42.24 dB gain margin.
- Verified transient response, PSRR, output noise, and load regulations through Cadence simulations.

POSITIONS OF RESPONSIBILITY

Teaching Assistant for **NPTEL : Basic Electrical Circuits**

Under Dr. Gajendranath Chaudhury | IIT Hyderabad.

EXTRACURRICULAR

NSS volunteer - JNTUK

LANGUAGE

English

Telugu

Hindi